OOP Final Project Report

需求分析

- 存储航班各种信息：

  实现 `Airline` 类，包含成员：

  - `std::string id_`：航班号；
  - `std::string start_`：始发站；
  - `std::string terminal_`：终点站；
  - `system_clock::time_point depart_time_`：起飞时间；
  - `system_clock::time_point arrive_time_`：到达时间；
  - `int ticket_count_`：初始票容量；
  - `int reserved_count_`：剩余票量。
- 从文件中读取航班信息：

  选取 `.csv` 格式的文件进行读写。
- 打印特定航班的详细信息（航班号，始终站，飞行时长，余票数量）：

  在航班类 `Airline` 中实现 `printInfo` 方法。
- 用户通过输入航班号/起飞时间/终点站查询特定航班：

  实际情况中用户查询特定起飞时间时，一般不会精确到小时，故使用起飞日期来查询。

  由于需要支持通过特定键值查询航班，故使用哈希表，例如 `std::unordered_map` 和 `__gnu_pbds::gp_hash_table`。由于后者效率更高，这里采用后者。

  使用 `std::vector<Airline>` 存储当前所有航班，并用三个 `__gnu_pbds::gp_hash_table` 存储特定航班号/起飞日期/终点站的航班在当前航班序列中的下标（不存储实例以节约空间）。

  由于多个航班的起飞日期/终点站可能相同，故后两者需要存储 `std::string, std::vector<int>` 键值对。在查询时，如有多个可能航班，将向用户打印这些航班并选择其一。
- 用户订票/退订：

  需要支持多个用户，并根据特定航班检查用户是否有预订信息，故需在 `Airline` 类中包含一个 `__gnu_pbds::gp_hash_table` 成员，用于存储“用户 id + 订票数量”键值对。

模块设计

核心类设计

`Airline` 类

封装航班相关的数据和操作。

成员变量：

```
private:
    std::string id_; // 航班号
    std::string start_; // 始发站
    std::string terminal_; // 终到站
    system_clock::time_point depart_time_; // 起飞时间
    system_clock::time_point arrive_time_; // 到达时间
    int ticket_count_; // 初始票数量
    int reserved_count_; // 已预订票数量
    __gnu_pbds::gp_hash_table<std::string, int> reserved_users_; // 当前预订此航班的客户
id 以及数量
```

成员函数:

```
public:
    explicit Airline(std::string id = "$",
        std::string start = "$",
        std::string terminal = "$",
        std::string depart = "1970-01-01 00:00",
        std::string arrive = "1970-01-01 00:00",
        int ticket_count = -1);
    ~Airline();
    std::string id() const;
    std::string start() const;
    std::string terminal() const;
    system_clock::time_point depart_time() const;
    system_clock::time_point arrive_time() const;
    int ticket_count() const; // 当前余票
    int initial_ticket_count() const; // 初始总票数
    const __gnu_pbds::gp_hash_table<std::string, int>& reserved_users() const; // 预订
用户信息
    minutes calcDuration(); // 计算飞行时长并以分钟返回
    int addReservation(std::string user_id, int count); // 为该航班添加一位客户的预订
    int delReservation(std::string user_id, int count); // 为该航班取消一位客户的预订
    void printTime(const system_clock::time_point& tp); // 打印 timepoint
    void printInfo(); // 打印信息: 1.航班号 2.起点/终点 3.起飞/到达时间 4.飞行时长 5.余票
```

`Solver` 类

用于实现与用户交互并维护航班信息。

成员变量:

```
private:
    Airline null_flight; // 用于查找失败时返回的默认航班
    std::vector<Airline> flights_; // 航班序列
    __gnu_pbds::gp_hash_table<std::string, int> ht_id_; // 用于按 id 查找航班在航班序列中的下标
    __gnu_pbds::gp_hash_table<std::string, std::vector<int>> ht_depart_; // 按出发时间查找，此处按日期分类，之后会列出本日所有航班供用户选择
    __gnu_pbds::gp_hash_table<std::string, std::vector<int>> ht_terminal_; // 按终点站查找
    // 由于多个航班出发时间和终点站可能相同，需要使用 vector 存储下标序列
```

成员函数：

```
public:
    Solver();
    ~Solver();
    void addFlight(); // 通过控制台输入添加一个新航班
    Airline& queryFlight(const std::string& key, int type); // 通过 id/出发时间/终点站 获取航班
    Airline& queryFlightInteract(); // 用户获取航班
    void reserveTicket(const std::string& user_id, int count, Airline& flight); // 向航班中添加用户订票信息
    void cancelTicket(const std::string& user_id, int count, Airline& flight); // 向航班中添加用户取消订票信息
    void reserveFlight(Airline& flight); // 用户订票（已知航班）
    void cancelFlight(Airline& flight); // 用户取消订票（已知航班）
    void reserveFlight(); // 用户订票（未知航班）
    void cancelFlight(); // 用户取消订票（未知航班）
    void addFlightFromFile(const std::string& path); // 从文件中读取新增航班信息
    void saveFlightToFile(const std::string& path); // 保存航班信息到文件
    void startInteract(); // 开始交互
```

CSV 文件格式

见 README.md。

核心功能

处理时间

采用 C++ `chrono` 库，实现了 `timeToString`、`printTime` 等函数/方法，用于字符串和时间的类型转换。

CSV 文件读写

实现了以下函数用于用户通过文件输入航班信息/向文件中存储当前航班信息（包括预订）：

```
void addFlightFromFile(const std::string& path);
void saveFlightToFile(const std::string& path);
```

哈希表实现快速操作

采用 GNU pbds 库中的高效哈希表 `gp_hash_table`，以实现以下操作时间复杂度：

- 按 ID 查询航班：$O(1)$；
- 按起飞日期/终点站查询航班：$O(k)$，其中 k 为同一起飞日期/终点站航班数；
- 为某一用户预订/退订特定航班的若干张机票：$O(1)$。

程序测试分析

测试用例1（目录下的 `test1.csv`）

```
Flight ID,Departure Station,Terminal Station,Departure Time,Arrival Time,Initial
Ticket Count
FK7891,Beijing,Ningbo,2026-04-23 14:13,2026-04-23 16:23,180,
FK9178,Ningbo,Beijing,2026-04-23 22:26,2026-04-23 23:55,220,
```

用于测试基本功能：根据三种关键字查询航班、预订/退订航班、保存至本地 CSV 文件。

流程测试

```
[NOTICE] Welcome to the flight reservation system
[NOTICE] Operation Menu:
1. Add Flight by Input in Console
2. Add Flight from Local CSV File
3. Query Flight Information
4. Start Reservation
5. Cancel Reservation
6. Save Flight and Reservation Information to Local CSV File
7. Show Format Help
8. Show This Menu Again
9. Exit
[NOTICE] Operation Menu END
- Please input your choice (type 8 to show menu again):
```

启动 `main.exe` 后，正常打印操作菜单。

```
- Please input your choice (type 8 to show menu again): 2
- Please input CSV file path: tests/test1.csv
[OP] Flights added successfully from file: tests/test1.csv
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 1
- ID Format: XXYYYY, where Xs are alphas and Ys are numbers.
- Please input Flight ID: FK7891
[NOTICE] Found flight with ID FK7891
- Do you want to show the flight info? (y/n): y
[FLIGHT] Flight Information
- Flight ID: FK7891
- Departure Station: Beijing
- Terminal Station: Ningbo
- Departure Time: 2026-04-23 14:13
- Arrival Time: 2026-04-23 16:23
- Flight Duration: 2h10min
- Remaining Ticket: 180
- Do you want to reserve tickets for this flight? (y/n): n
- Do you want to cancel reservation for this flight? (y/n): n
- Please input your choice (type 8 to show menu again):
```

输入 `2` 从 `tests/test1.csv` 中读取了两个航班信息，并根据航班号对 FK7891 航班进行了查询。查询成功后，系统询问用户是否要展示该趟航班信息、是否要预订或取消预订。可以看到，`Airline` 类正确地根据起飞时间和到达时间计算了航班的飞行时长。

```
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 2
- Date format: YYYY-MM-DD
- Please input Departure Date: 2026-04-23
[NOTICE] There are multiple flights with Departure Date 2026-04-23, listed below:
- [FLIGHT] FK7891, precise time:  14:13
- [FLIGHT] FK9178, precise time:  22:26
- Please choose one of them with ID: FK9178
[NOTICE] Found flight with ID FK9178
- Do you want to show the flight info? (y/n): n
- Do you want to reserve tickets for this flight? (y/n): n
- Do you want to cancel reservation for this flight? (y/n): n
- Please input your choice (type 8 to show menu again):
```

输入 `3` 根据起飞日期查询航班。由于两趟航班均为 2026.4.23 起飞，故系统展示了两趟航班的起飞具体时间，并询问用户要查询哪一趟。

```
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 3
- Please input Terminal Station: Beijing
[NOTICE] Found flight with ID FK9178
- Do you want to show the flight info? (y/n): n
- Do you want to reserve tickets for this flight? (y/n): y
[OP] Reservation
- Please input your user ID: neutralized
- Please input ticket amount: 3
[OP] Reservation from user neutralized for flight FK9178 with 3 tickets succeeded.
- Do you want to cancel reservation for this flight? (y/n): y
[OP] Canceling Reservation
- Please input your user ID: neuze
- Please input ticket amount: 1
[ERROR] User neuze hasn't reserved any ticket for flight FK9178
- Please input your choice (type 8 to show menu again):
```

输入 `3` 根据终点站进行查询，随后选择预订 FK9178 号航班，用户为 `neutralized`，预订了 3 张机票；

选择对该航班进行退订，用户为 `neuze`，但之前并没有该用户的预订信息，所以系统返回了一个 ERROR。

```
- Please input your choice (type 8 to show menu again): 1
[ADD] Adding a flight
- Flight ID: MU1145
- Departure Station: Chengdu
- Terminal Station: Xi'An
- Departure Time: 2026-04-26 10:12
- Arrival Time: 2026-04-26 11:23
- Ticket Count: 300
- Please input your choice (type 8 to show menu again): 4
[OP] Reservation
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 3
- Please input Terminal Station: Xi'An
- Please input your user ID: neuze
- Please input ticket amount: 4
[OP] Reservation from user neuze for flight MU1145 with 4 tickets succeeded.
- Please input your choice (type 8 to show menu again):
```

输入 `1`，在控制台中手动输入新航班 MU1145 的信息，并根据终点站查询了该航班，随后用户 `neuze` 预订了 4 张机票。

```
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 1
- ID Format: XXYYYY, where Xs are alphas and Ys are numbers.
- Please input Flight ID: MU1145
[NOTICE] Found flight with ID MU1145
- Do you want to show the flight info? (y/n): y
[FLIGHT] Flight Information
- Flight ID: MU1145
- Departure Station: Chengdu
- Terminal Station: Xi'An
- Departure Time: 2026-04-26 10:12
- Arrival Time: 2026-04-26 11:23
- Flight Duration: 1h11min
- Remaining Ticket: 296
- Do you want to reserve tickets for this flight? (y/n): n
- Do you want to cancel reservation for this flight? (y/n): n
- Please input your choice (type 8 to show menu again):
```

输入 `3`，对刚才的航班进行了查询。由于刚才的预订，此时余票数量变为了 296。尝试对该航班进行退订，再次查询余票数量：

```
- Please input your choice (type 8 to show menu again): 5
[OP] Canceling Reservation
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 2
- Date format: YYYY-MM-DD
- Please input Departure Date: 2026-04-26
- Please input your user ID: neuze
- Please input ticket amount: 5
[OP] Cancelation from user neuze for flight MU1145 with 5 tickets succeeded.
[NOTICE] User canceled with a ticket count larger than reservation, so system cleared all reservation from user
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 1
- ID Format: XXYYYY, where Xs are alphas and Ys are numbers.
- Please input Flight ID: MU1145
[NOTICE] Found flight with ID MU1145
- Do you want to show the flight info? (y/n): y
[FLIGHT] Flight Information
- Flight ID: MU1145
- Departure Station: Chengdu
- Terminal Station: Xi'An
- Departure Time: 2026-04-26 10:12
- Arrival Time: 2026-04-26 11:23
- Flight Duration: 1h11min
- Remaining Ticket: 300
- Do you want to reserve tickets for this flight? (y/n): n
- Do you want to cancel reservation for this flight? (y/n): n
- Please input your choice (type 8 to show menu again):
```

此时余票数量回到了 300。同时，由于 `neuze` 尝试取消预订 5 张机票，但他只预订过 4 张，所以系统发出了提醒，并且把 4 张机票全部取消了预订。

```
- Please input your choice (type 8 to show menu again): 6
- Please input saving path: tests/output.csv
[OP] Flights saved successfully to file: tests/output.csv
- Please input your choice (type 8 to show menu again):
```

最后，用户让系统将目前的航班信息输出到指定文件中，然后退出了程序。

异常测试

输入不存在的选项：

```
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 4
[ERROR] Input format error: option not included
[ERROR] Flight query failed
- Please input your choice (type 8 to show menu again):
```

航班号格式错误：

```
- Please input your choice (type 8 to show menu again): 1
[ADD] Adding a flight
- Flight ID: f
[ERROR] Flight ID format error
- Please input your choice (type 8 to show menu again):
```

起飞/到达时间格式错误：

```
- Please input your choice (type 8 to show menu again): 4
[OP] Reservation
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 2
- Date format: YYYY-MM-DD
- Please input Departure Date: 2026.4.14
[ERROR] Date format error: not formated correctly
[ERROR] Flight query failed
- Please input your choice (type 8 to show menu again):
```

查询失败：

```
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 3
- Please input Terminal Station: New Jersey
[ERROR] Flights with Terminal Station New Jersey doesn't exist
[ERROR] Flight query failed
```

测试用例2（用 `tests` 目录下的 `generate.bat` 生成）

用于测试大规模输入/输出。

`generate.bat` 运行结果：

```
==================================
Test Data Generator
==================================

[INFO] Generating test2.csv with 10,000 flight records...
[SUCCESS] test2.csv generated successfully!
[INFO] File size: 622220 bytes

==================================
Generation completed!
==================================
请按任意键继续. . .
```

为保证 CSV 文件被正确读入，我们选取文件靠后的一趟航班进行查询：

```
UF6912,Ningbo,Guangzhou,2026-01-08 13:43,2026-01-08 18:49,245
SU2206,Shenzhen,Beijing,2026-04-14 08:07,2026-04-14 23:05,108
ZL8121,Chengdu,Beijing,2026-07-27 00:03,2026-07-27 10:32,330
```

```
- Please input your choice (type 8 to show menu again): 3
- Querying Flights:
- Input 1 for ID, 2 for Departure Date, 3 for Terminal Station: 2
- Date format: YYYY-MM-DD
- Please input Departure Date: 2026-04-14
[NOTICE] There are multiple flights with Departure Date 2026-04-14, listed below:
- [FLIGHT] CT9522, precise time:  18:31
- [FLIGHT] IL8678, precise time:  06:56
- [FLIGHT] WT0364, precise time:  11:47
- [FLIGHT] ZQ1377, precise time:  11:26
- [FLIGHT] CI4438, precise time:  20:28
- [FLIGHT] AM4128, precise time:  17:44
- [FLIGHT] PC3833, precise time:  22:34
- [FLIGHT] AJ8873, precise time:  16:43
- [FLIGHT] IA1633, precise time:  07:03
- [FLIGHT] YW0522, precise time:  16:26
- [FLIGHT] VS4634, precise time:  00:39
- [FLIGHT] GM0092, precise time:  11:31
- [FLIGHT] KN0620, precise time:  15:07
- [FLIGHT] AY0385, precise time:  05:47
- [FLIGHT] BI9770, precise time:  23:06
- [FLIGHT] BP6747, precise time:  01:38
- [FLIGHT] DQ3732, precise time:  19:28
- [FLIGHT] AF7377, precise time:  09:12
- [FLIGHT] HQ9027, precise time:  23:52
- [FLIGHT] AJ0750, precise time:  06:30
- [FLIGHT] TV4784, precise time:  07:51
- [FLIGHT] GC2207, precise time:  01:48
- [FLIGHT] GF3110, precise time:  02:22
- [FLIGHT] SU2206, precise time:  08:07
- [FLIGHT] CD6010, precise time:  01:49
- [FLIGHT] FC0425, precise time:  13:06
- [FLIGHT] CA0729, precise time:  11:25
- [FLIGHT] HI1810, precise time:  15:33
- [FLIGHT] CO6046, precise time:  07:59
- [FLIGHT] ZP2821, precise time:  04:21
- [FLIGHT] SM4431, precise time:  09:23
- [FLIGHT] EY3909, precise time:  16:10
- [FLIGHT] SF2012, precise time:  10:00
- [FLIGHT] OJ2107, precise time:  10:32
- [FLIGHT] QC8756, precise time:  22:45
- Please choose one of them with ID:
```

SU2206 号航班位于图中靠后的位置。此时输入 ID 进行二次查询，并尝试预订 114 张机票，系统返回了一个错误，原因是该航班只有 108 的票容量，无法进行这次预订。

```
- Please choose one of them with ID: SU2206
[NOTICE] Found flight with ID SU2206
- Do you want to show the flight info? (y/n): y
[FLIGHT] Flight Information
- Flight ID: SU2206
- Departure Station: Shenzhen
- Terminal Station: Beijing
- Departure Time: 2026-04-14 08:07
- Arrival Time: 2026-04-14 23:05
- Flight Duration: 14h58min
- Remaining Ticket: 108
- Do you want to reserve tickets for this flight? (y/n): y
[OP] Reservation
- Please input your user ID: Neutralized
- Please input ticket amount: 114
[ERROR] Ticket of flight SU2206 is not enough for reservation request!
- Do you want to cancel reservation for this flight? (y/n):
```

更多运行截图限于篇幅不在此展示。可以得出结论，程序在 $n = 10000$ 规模的随机数据上运行效果良好。

结论

程序在小规模和较大规模数据上均运行良好。